$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}'_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] - 2M\gamma_0 k_{\mathrm{B}} T\big[\hat{X}, \big[\hat{X}, \hat{\rho}_{\mathcal{S}}(t)\big]\big], \tag{5.62}$$

or, expressed in the position basis,

$$\frac{\partial}{\partial t}\rho_{\mathcal{S}}(X, X', t) = \Bigg[-\frac{\mathrm{i}}{2M}\left(\frac{\partial^2}{\partial X'^2} - \frac{\partial^2}{\partial X^2}\right) - \frac{\mathrm{i}}{2}M\left(\Omega^2 - 2\gamma_0\Lambda\right)\left(X^2 - X'^2\right) - \gamma_0\left(\frac{X - X'}{\lambda_{\mathrm{dB}}}\right)^2\Bigg]\rho_{\mathcal{S}}(X, X', t). \tag{5.63}$$